

Predicting Recognized Developmental Concern or Support in Preschool Children

A machine-learning proof of concept using the 2024 U.S. National Survey of Children's Health

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7,485

children ages 3–5

27.1%

positive composite outcome

44

features in reduced Model C

1. Problem framing

Healthcare classification problem with a deliberately non-diagnostic target

Research question

Can supervised machine learning identify children ages 3–5 who meet a composite outcome of recognized developmental concern or support?

Secondary question

Can a smaller, more interpretable feature set retain comparable predictive performance?

Task

Supervised binary classification

Target

Recognized developmental concern or support

Primary metrics

ROC-AUC • PR-AUC • F1 • sensitivity • precision • specificity

Guardrail

This target reflects recognition and support, not a confirmed diagnosis or latent developmental need.

Why it matters

Future use would be decision support for further review, not autonomous clinical decision-making.

2. Dataset and outcome

2024 National Survey of Children's Health, public-use data

51,375

rows in source dataset

7,485

children ages 3–5

102

candidate predictors

27.1%

positive outcome

Composite outcome: positive if ≥ 1 qualifying indicator

- Relevant developmental / behavioral / learning / communication diagnosis
- Developmental, behavioral, or mental-health treatment or service
- Early-intervention or special-education support
- Other qualifying developmental support indicators in the survey

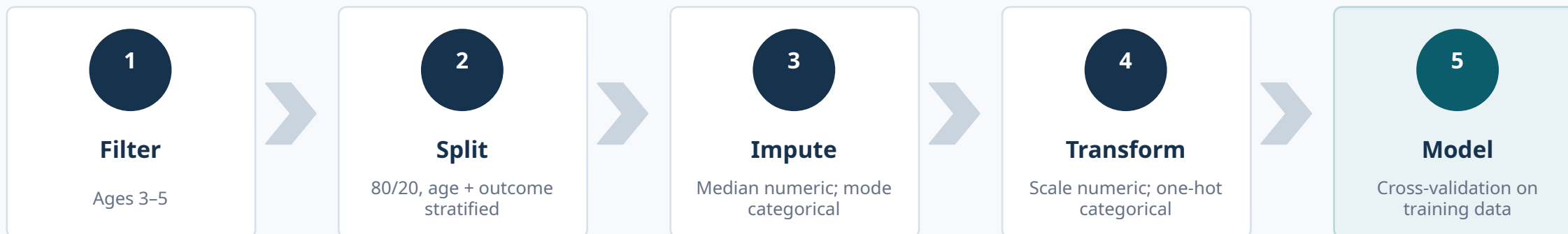
Data quality snapshot

- No exact duplicate rows
- Maximum predictor missingness: 4.07%
- No predictor >10% missingness
- 4 variables treated as numeric; 98 as categorical
- Age + outcome stratification used for train / test split

Source: 2024 U.S. National Survey of Children's Health Topical Public Use File.

3. Preprocessing and applied EDA

All transformations were embedded in modeling pipelines to reduce leakage



Applied EDA

- Outcome class balance
- Age distribution and age-by-outcome strata
- Missingness and feature distributions
- Subgroup performance checks

Dimensionality reduction

PCA showed substantial overlap between outcome groups in the first two components. The composite outcome is not a simple low-dimensional phenotype.

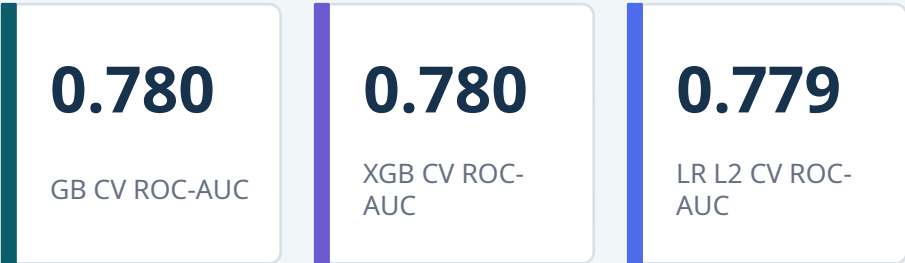
4. Modeling strategy

Broad benchmark first, then tune finalists and interrogate model behavior

Initial 5-fold CV benchmark

Model	ROC-AUC	PR-AUC
Logistic L1	0.774	0.653
Logistic L2	0.771	0.651
Linear SVM	0.770	0.649
Random Forest	0.762	0.632
Gradient Boosting	0.776	0.656
XGBoost	0.742	0.610
KNN	0.625	0.413

Finalists after tuning



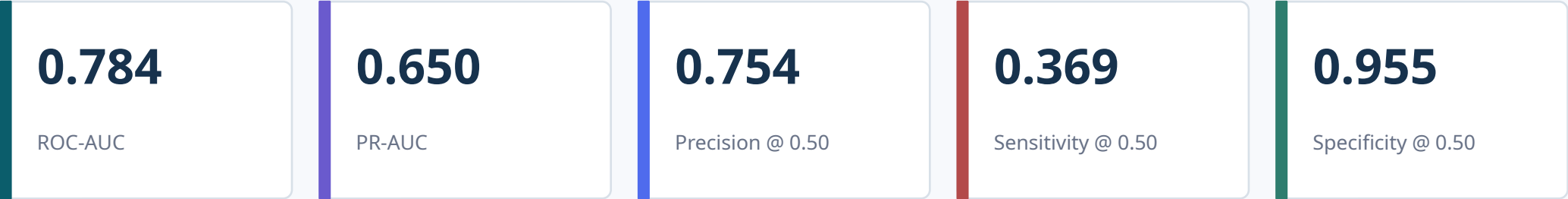
Gradient Boosting was retained as the reference model because it combined strong discrimination, slightly better downstream calibration, and convenient tree-based explainability.

Cross-validation used training data only. Hyperparameters were tuned before final model comparisons.

5. Model A: full-feature performance

102 predictors captured useful signal, but the default threshold was too conservative for a screening-oriented use case

Held-out test performance | Gradient Boosting



Threshold trade-off



Operating threshold is a workflow decision, not a diagnostic cutoff.

Calibration

Brier score **0.1478**

Predicted probabilities were reasonably calibrated internally, but external calibration has not been established.

6. Feature reduction experiment

Can we preserve performance with a smaller, more interpretable predictor set?

Model A

102

broad eligible predictors

- Developmental
- Medical
- Family / caregiver
- ACEs
- Routines / environment

Model B

38

domain-informed predictors

- Caregiver-observable
- Development-focused
- Clinically coherent
- Practical for future intake

Model C

44

data-driven SHAP-ranked predictors

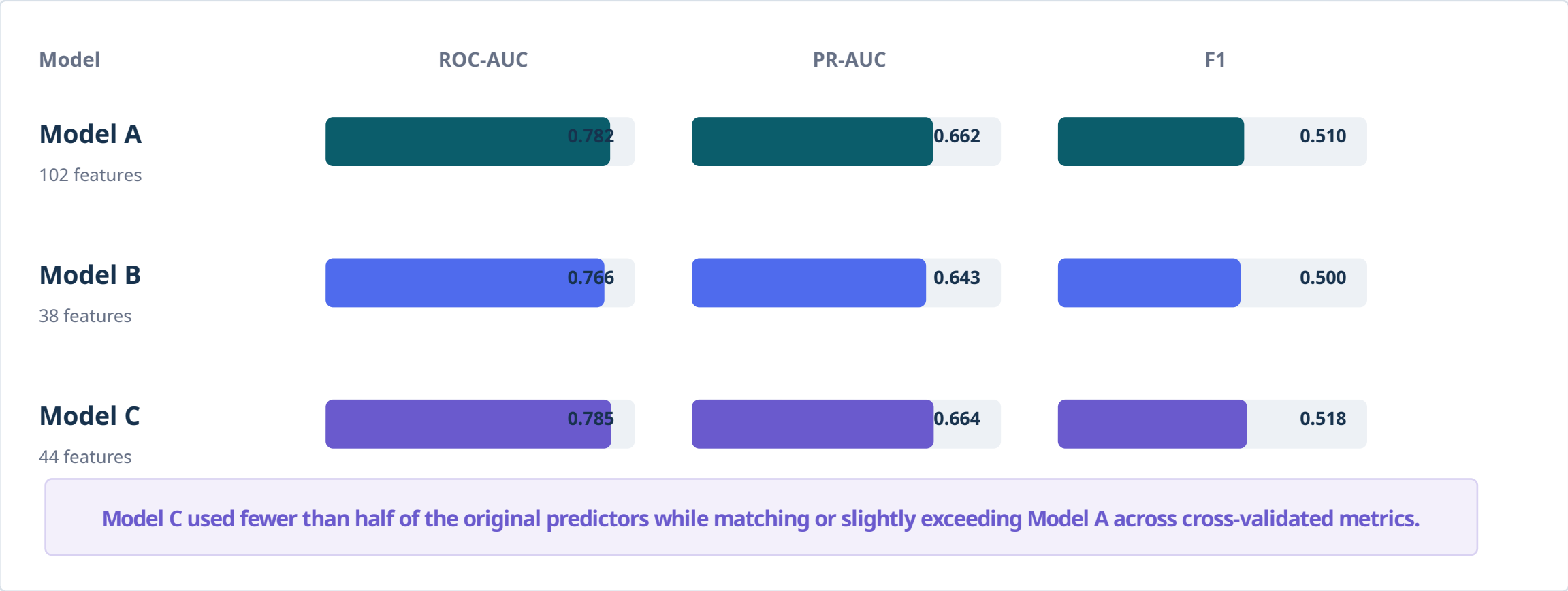
- Rank Model A by SHAP importance
- Evaluate top 20, 22, 24...
- Find performance plateau
- Retain smallest near-plateau set

22 predictors were shared between Models B and C

Model C selection was exploratory because SHAP ranking and subset evaluation both used the training dataset. Nested selection would be preferred for confirmatory research.

7. Final apples-to-apples comparison

Same Gradient Boosting architecture, same 5-fold CV, different feature sets

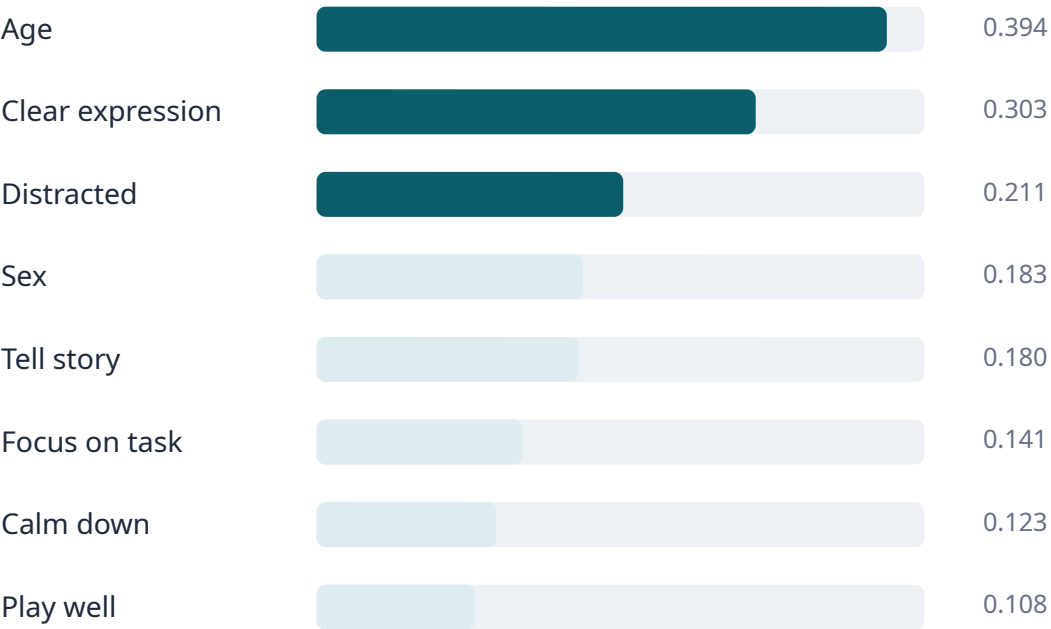


The small apparent advantage of Model C should be interpreted cautiously because feature ranking was exploratory and not nested within cross-validation.

8. Explainability and fairness audit

Predictive performance was not treated as sufficient evidence of responsible use

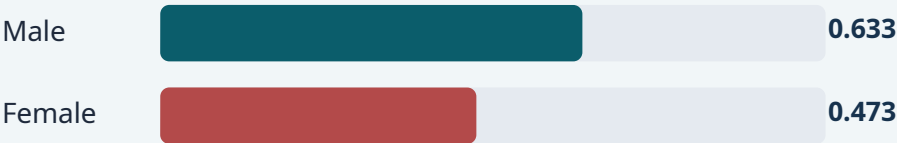
Top SHAP predictors | Model B



SHAP indicates predictive contribution, not causation.

Sex-based fairness audit @ threshold 0.30

Sensitivity



Flagged ratio: Male 36.3% | Female 19.2%

Key ethical concern

Because the target reflects recognized concern/support, the labels themselves may encode historical under-recognition. A model can learn and reproduce that pattern.

9. Conclusions and next research steps

What this capstone supports, and what it does not

What the analysis supports

- Machine learning can identify meaningful patterns associated with recognized developmental concern/support.
- A reduced 44-feature model retained the predictive signal of the 102-feature model.
- Domain-informed and data-driven feature selection showed substantial convergence.
- Threshold choice materially changes who is flagged.
- Fairness auditing reveals risks hidden by aggregate performance.

What comes next

- 1 Nested feature selection / independent validation
- 2 Prospective outcome labels closer to developmental need
- 3 Explicit fairness mitigation and subgroup monitoring
- 4 Local Philippine validation and recalibration
- 5 Human-in-the-loop workflow testing before deployment

Bottom line: promising proof of concept, not a clinically deployable screening instrument.